

UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II
WEB TECHNOLOGIES – LECTURE 03

CSS: CASCADING STYLE SHEETS

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PREVIOUSLY, ON WEB TECHNOLOGIES

- We have learned how to write HTML documents
- HTML is concerned with **structure** and **semantics** of documents
- HTML is saying nothing at all on the **appearance** of documents

- An **** element specifies that its content should be emphasized
- It's not saying **how** the emphasizing part should be done
 - Emphasis might be conveyed using *italics*, **different colors** or **backgrounds**.

CSS: CASCADING STYLE SHEETS

- A **rule-based, declarative** language for specifying how documents should be presented to users.
- A **stylesheet** is a set of **Rules**, each defined as follows
- The **selector** specifies which HTML elems are affected by the rule
- Rules contain a set of **declarations**, in the form of **property-value pairs**, which specify the style to apply

```
selector {  
  property: value;  
  property: value;  
}
```

CSS: FIRST EXAMPLE

```
<h1>Hello CSS</h1>
<p>
  Our <em>first</em> page
  with <em>style</em>!
</p>
```

```
h1 {
  color: red;
  font-size: 50px;
}

em {
  color: blue;
}
```



DEFAULT USER AGENT STYLES

- But the first web pages we developed have some styling!
 - Headings are bigger and shown with a bold face...
 - `<p>` starts on new lines, `<em>` are displayed in italic, `<strong>` in bold, `<a>` are underlined and blue, `<ul>` have bullets, and so on...
- That's because browsers apply their own, basic styles to every page!
- They are often referred to as **user agent styles**
- These defaults are roughly the same across different browsers, but some **differences** exist (and we'll get back to that!)

INCLUDING STYLESHEETS IN WEB PAGES

Styling can be included in HTML documents in different ways

- Using `<link>` elements in the `<head>` of the document
 - The `rel="stylesheet"` attribute specifies the relation between the current document and the linked document
 - The `href="style.css"` attrib. specifies the URL of the stylesheet to load
 - Same mechanism as `<img>`: browser will make an additional HTTP request to fetch the stylesheet before rendering the page

```
<head>  
  <meta charset="UTF-8">  
  <title>CSS</title>  
  <link rel="stylesheet" href="style.css">  
</head>
```

INCLUDING STYLESHEETS IN WEB PAGES

- CSS rules can also be defined in `<style>` elements in the `<head>`
- It is generally preferable to use external stylesheets and `<link>`
 - Can you think of some reasons why?

```
<head>
  <meta charset="UTF-8">
  <title>CSS</title>
  <style>
    h1 {
      color: red;
      font-size: 50px;
    }
  </style>
</head>
```

STYLING HTML ELEMENTS

- HTML elements can also be styled inline, using the **style** attribute
- The value of the style attribute is a sequence of declarations, separated by « ; »
- These styling declarations apply **only to the specific element** bearing the attribute

```
<em style="color: fuchsia; font-weight: bold;">inline style</em>
```

CSS: INLINE STYLES

```
<h1>Hello CSS</h1>
<p>
  Our <em style="color:fuchsia;font-weight: bold;">first</em>
  page with <em>style</em>!
</p>
```

```
h1 {
  color: red;
  font-size: 50px;
}

em {
  color: blue;
}
```



SELECTORS



SELECTORS

- Selectors are a **key** part of CSS
- They specify to which elements a CSS rule applies
- CSS selectors are not only used for styling!
 - When using JavaScript to make web pages dynamic, they can be used to select which elements to interact with
 - When doing automated web testing, they can be used to determine which elements the test needs to interact with
 - When doing scraping/crawling, they can be used to select the elements that contain the information we want to extract

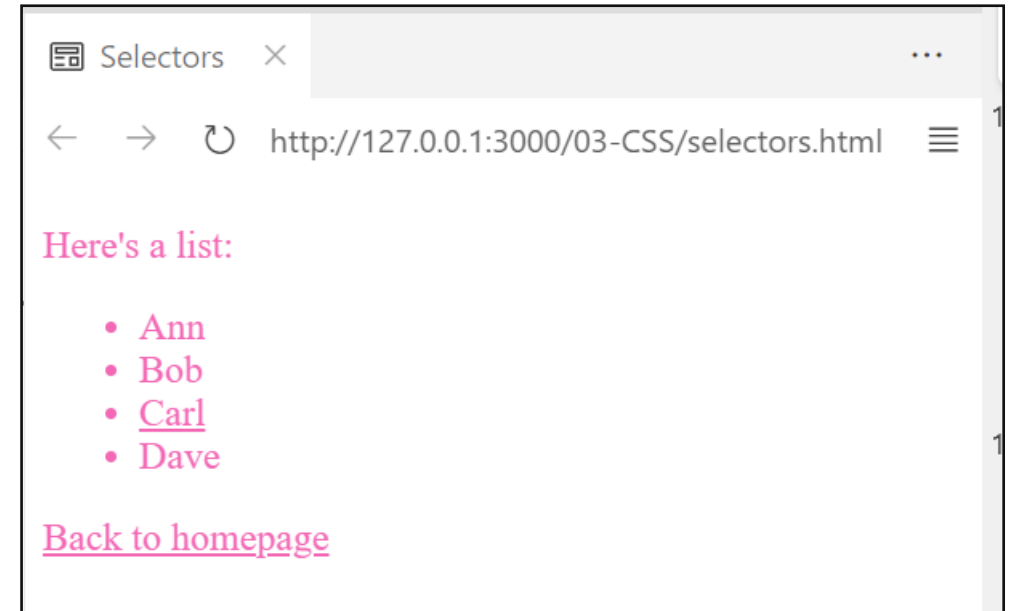
SIMPLE CSS SELECTORS

There exist **five** kinds of simple selectors:

- **Universal** selector (a.k.a **wildcard**). Matches any element.

```
* {  
  color: hotpink;  
}
```

```
<p>Here's a list:</p>  
<ul>  
  <li>Ann</li>  
  <li>Bob</li>  
  <li><a href="/car/">Carl</a></li>  
  <li>Dave</li>  
</ul>  
<a href="/">Back to homepage</a>
```

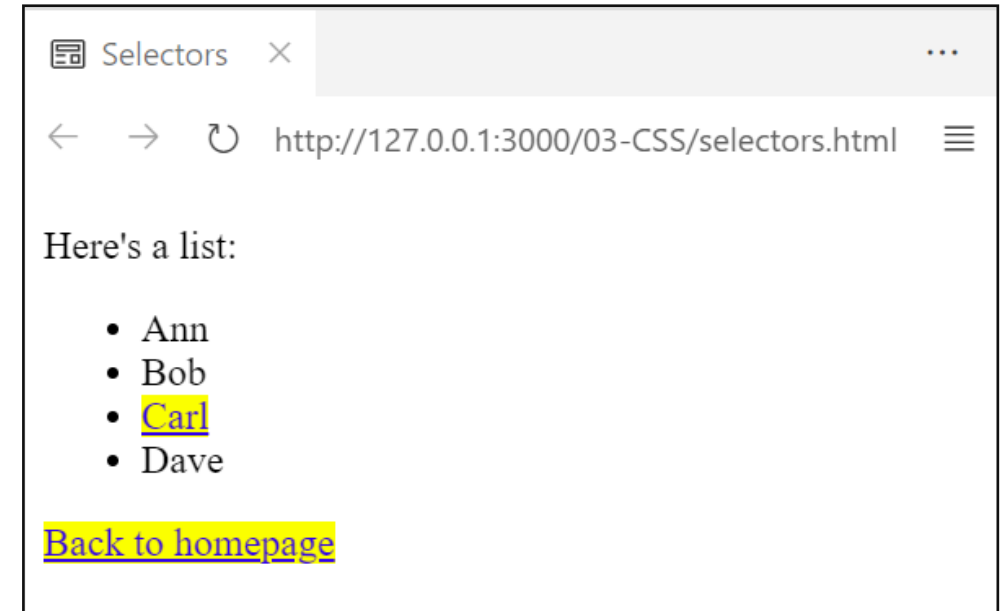


SIMPLE CSS SELECTORS

- **Type** selector. Matches all element of a given type (i.e., tag name)
- The selector is simply the name of the tag to match

```
a {  
  background: yellow;  
}
```

```
<p>Here's a list:</p>  
<ul>  
  <li>Ann</li>  
  <li>Bob</li>  
  <li><a href="/car/">Carl</a></li>  
  <li>Dave</li>  
</ul>  
<a href="/">Back to homepage</a>
```

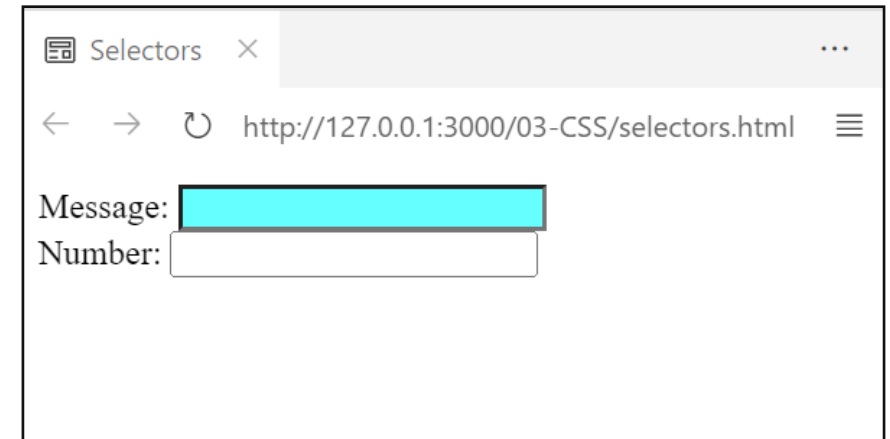


SIMPLE CSS SELECTORS

- **Id** selector. Matches the element with the given **id** attribute.
- Selector has the form **#*ElementId***

```
#msg {  
  background: cyan;  
}
```

```
<form>  
  <label for="msg">Message: </label>  
  <input id="msg" type="text" name="msg"><br>  
  <label for="num">Number: </label>  
  <input id="num" type="number" name="num">  
</form>
```



SIMPLE CSS SELECTORS

- **Class selector.** Matches the element with the given **class** attribute.
- Selector has the form **.classname**

```
.primary {  
  background: blue;  
  color: white;  
}
```

```
<button>Reset form</button>  
<button class="primary btn">Continue</button>
```

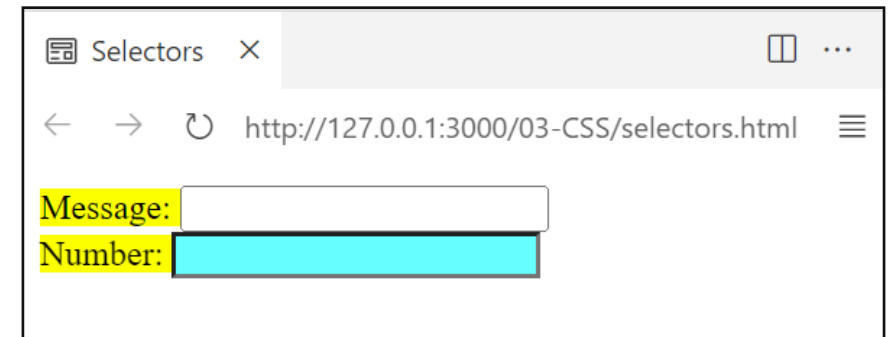


SIMPLE CSS SELECTORS

- **Attribute** selector. Matches the element with a certain attribute.
- Selector has the form **[attribute]** or **[attribute='value']**

```
[for]{ /*all elems with a for attribute*/  
  background: yellow;  
}  
[type='number']{ /*all elems with type=number*/  
  background: cyan;  
}
```

```
<form>  
  <label for="msg">Message: </label>  
  <input id="msg" type="text" name="msg"><br>  
  <label for="num">Number: </label>  
  <input id="num" type="number" name="num">  
</form>
```



SIMPLE CSS SELECTORS

- Additional operators (`*=`, `^=`, `$=`) allow **partial matching** with attribute values

```
[href*='programming']{ /*contains 'programming'*/  
  text-decoration: underline;  
}  
[href^='https']{ /*start with 'https'*/  
  color: red;  
}  
[href$='.it/']{ /*ends with '.it/'*/  
  color: green;  
}
```

```
<a href="http://bookofprogramming.com/">Link 1</a>  
<a href="https://programming.net/">Link 2</a>  
<a href="http://webtechnologies.it/">Link 3</a>
```



COMPLEX CSS SELECTORS: COMPOUNDS

- It is possible to combine selectors to get fine-grained control
- This is done by concatenating selectors
- Basically select the **intersection** of the involved selectors

```
a[target='_blank'] {  
  color: red;  
}  
a.my-class{  
  color: green;  
}  
a[href*='programming'].my-class {  
  background: yellow;  
}
```

```
<a href="http://bookofprogramming.com/" target="_blank">Link 1</a>  
<a class="my-class" href="https://programming.net/">Link 2</a>  
<em class="my-class">Hello</em>
```



COMPLEX CSS SELECTORS: COMBINATORS

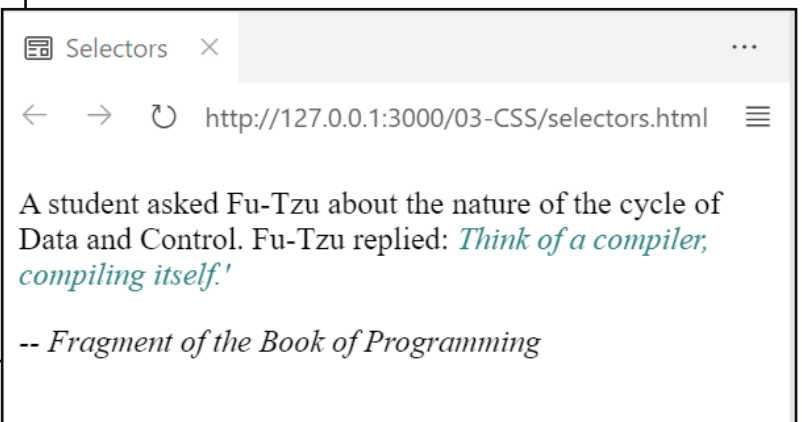
- Combinators are used to select elements based on their position in the document (remember that HTML documents can be seen as **trees**!)
- Syntax is: `selector1 combinator selector2`
- Four different combinators exist in CSS:
 - Descendant selector (space)
 - Child selector (`>`)
 - Adjacent sibling selector (`+`)
 - General sibling selector (`~`)

COMBINATORS: DESCENDANT SELECTOR

- Syntax: selectorA selectorB
- Semantics: match all elements that match selectorB and are a contained within (i.e., are a descendant of) an element matching selectorA

```
<section>
  <p>
    A student asked Fu-Tzu about the nature of
    the cycle of Data and Control. Fu-Tzu replied:
    <em>Think of a compiler, compiling itself.</em>
  </p>
</section>
<em>-- Fragment of the Book of Programming</em>
```

```
section em {
  color: teal;
}
```



COMBINATORS: CHILD SELECTOR

- Syntax: selectorA > selectorB
- Semantics: match all elements that match selectorB and are a **directly** contained within (i.e., are a direct child of) an element matching selectorA

```
main > em {  
  color: teal;  
  font-variant: small-caps;  
}
```

```
<main>  
  CSS <em>selectors</em>:  
  <p>We <em>like</em> 'em.</p>  
</main>
```

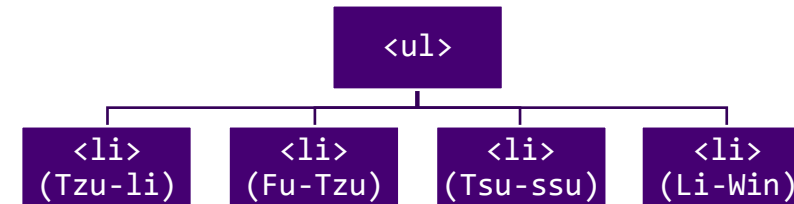


COMBINATORS: ADJACENT SIBLINGS

- Syntax: selectorA + selectorB
- Semantics: match all elements that match selectorB and are a **next adjacent siblings** of an element matching selectorA

```
.master + li {  
  color:red;  
}
```

```
<ul>  
  <li>Tsu-li</li>  
  <li class="master">Fu-Tzu</li>  
  <li>Tsu-ssu</li>  
  <li class="disciple">Li-Win</li>  
</ul>
```

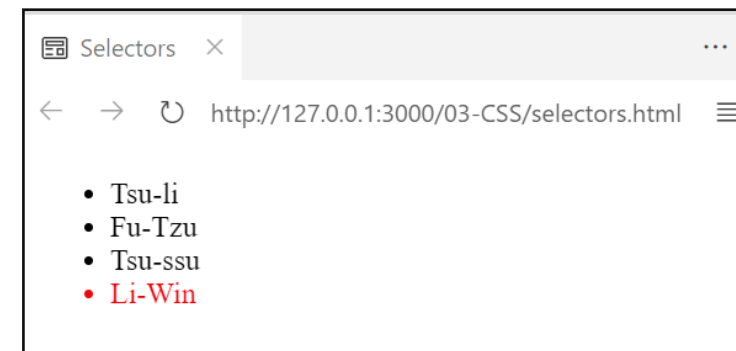
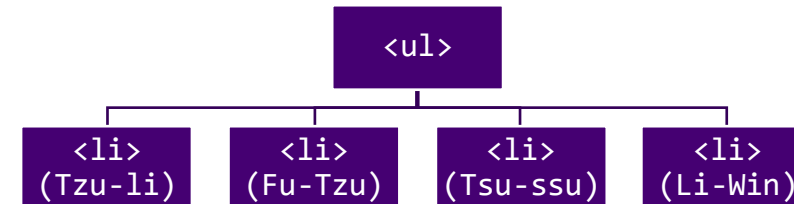


COMBINATORS: GENERAL SIBLINGS

- Syntax: `selectorA ~ selectorB`
- Semantics: match all elements that match `selectorB` and are a **subsequent siblings** of an element matching `selectorA`

```
.master ~ li.disciple {  
  color:red;  
}
```

```
<ul>  
  <li>Tsu-li</li>  
  <li class="master">Fu-Tzu</li>  
  <li>Tsu-ssu</li>  
  <li class="disciple">Li-Win</li>  
</ul>
```



CSS SELECTORS: PSEUDO-CLASSES

HTML elements can be in different **states**, for example because of **user interactions** or because of their relation with other elements.

Pseudo-classes selectors start with « : », and allow to style elements based on their **state**:

- **Interactive states** (resulting from user interaction)
- **Historic states** (used to «remember» which links were visited)
- **Form states** (specific of interaction with forms)
- States deriving from **relations with other elements**

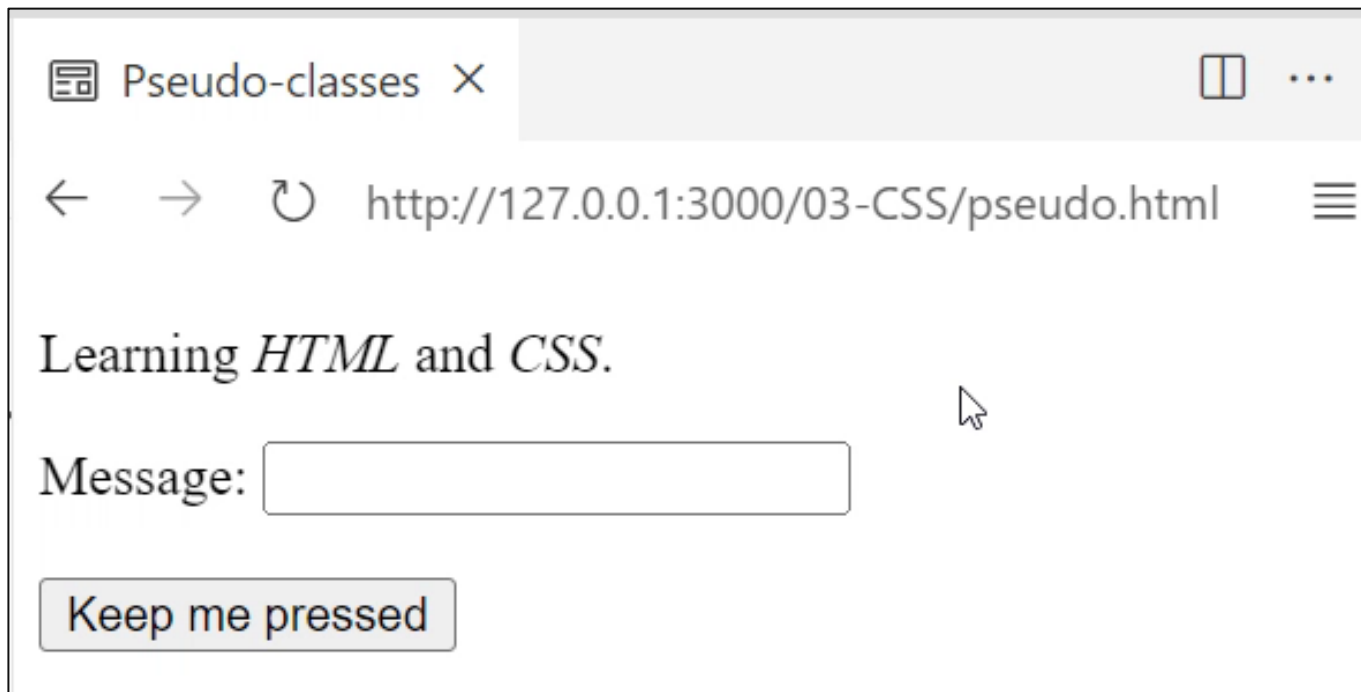
PSEUDO-CLASSES: INTERACTIVE STATES

- **:hover** selects the elements on which a pointing device (i.e.: mouse) is placed over
- **:active** matches the state in which an element is actively being interacted with (e.g.: button is being pressed)
- **:focus** matches the state in which an element (e.g.: a link or an input field) has focus (i.e.: is currently selected) in the web page

PSEUDO-CLASSES: INTERACTIVE STATES

```
<p>Learning <em>HTML</em> and <em>CSS</em>.</p>  
Message: <input type="text"><br><br>  
<button>Keep me pressed</button>
```

```
em:hover {  
    background: yellow;  
}  
input[type='text']:focus {  
    background: cyan;  
}  
button:active {  
    background: blue;  
    color: white;  
}
```

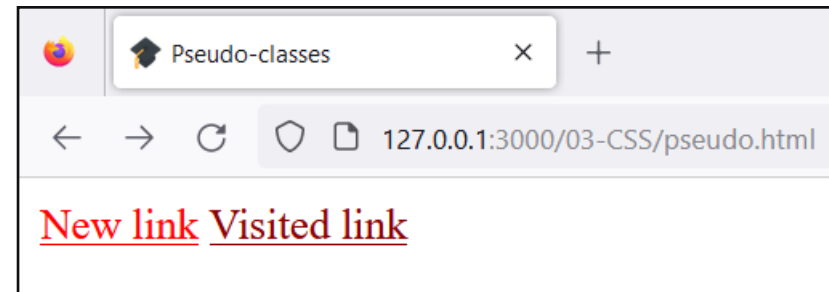


PSEUDO-CLASSES: HISTORIC STATES

- **:link** selects links that have not been visited yet
- **:visited** selects links that have already been visited

```
:link{  
  color: red;  
}  
:visited{  
  color: darkred;  
}
```

```
<a href="./js/">New link</a>  
<a href="./css/">Visited link</a>
```

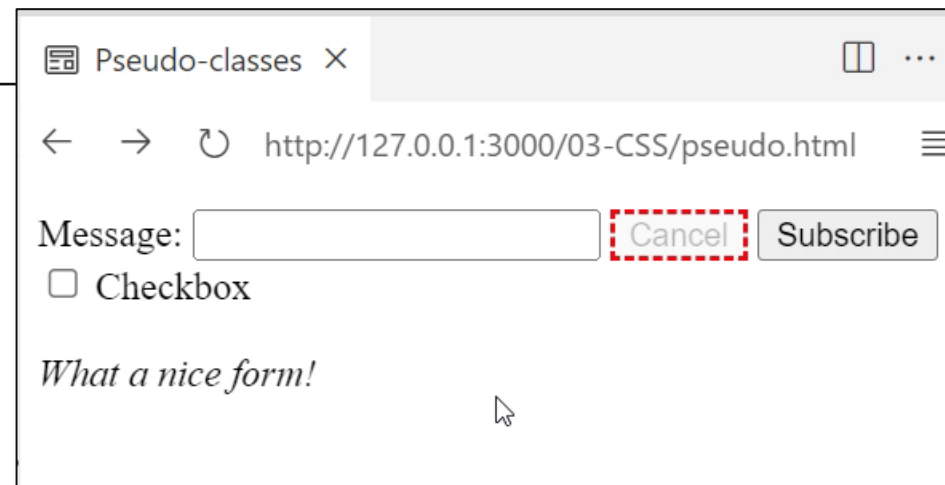


Historic states are **not supported** in the *Live Preview* browser within Visual Studio Code! Open the page in Firefox to check them out.

PSEUDO-CLASSES: FORM STATES

```
<div>
  <label for="mail">Message:</label><input id="mail" type="email">
  <button disabled>Cancel</button><button>Subscribe</button>
</div>
<input type="checkbox"> Checkbox<br><br>
<em>What a nice form!</em>
```

```
:disabled {
  border: 2px dashed red;
}
:invalid {
  color: red;
}
:checked ~ em {
  color: deeppink; font-weight: bold;
}
```



Technically, there can be email address without a dot. For example, user@localhost or user@com are valid addresses!

PSEUDO-CLASSES: POSITION RELATIONS

- **:first-child** and **:last-child** select the first (last) child among a set of siblings, independently from the tag.
- **:only-child** can be used to select elements that have no siblings.
- **:first-of-type** and **:last-of-type** can be used to select elements that are the first (last) child among a set of siblings, considering only elements of the same type.
- **:nth-child(n)** and **:nth-of-type(n)** can be used to select elements that are in the n-th position among their siblings.
 - Indexing in CSS starts at 1!

PSEUDO-CLASSES: POSITION RELATIONS

```
<p>  
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>  
</p>  
<p>  
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>  
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

PSEUDO-CLASSES: POSITION RELATIONS

```
<p>  
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>  
</p>  
<p>  
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>  
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

```
em:last-child {  
  color: red;  
}
```

Ann **Bob** Carl
Ann Bob **Car**

PSEUDO-CLASSES: POSITION RELATIONS

```
<p>  
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>  
</p>  
<p>  
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>  
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

```
em:last-of-type {  
  color: red;  
}
```

Ann **Bob** Carl
Ann **Bob** Car

PSEUDO-CLASSES: POSITION RELATIONS

```
<p>  
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>  
</p>  
<p>  
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>  
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

```
em:first-child {  
  color: red;  
}
```

Ann Bob Carl
Ann Bob **Car**

PSEUDO-CLASSES: POSITION RELATIONS

```
<p>  
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>  
</p>  
<p>  
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>  
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

```
em:first-of-type {  
  color: red;  
}
```

Ann Bob Carl
Ann **Bob** Car

PSEUDO-CLASSES: POSITION RELATIONS

```
<p>  
  <em>Ann</em> <strong>Bob</strong> <em>Carl</em>  
</p>  
<p>  
  <strong>Ann</strong> <em>Bob</em> <strong>Carl</strong>  
</p>
```

Ann **Bob** Carl
Ann Bob **Car**

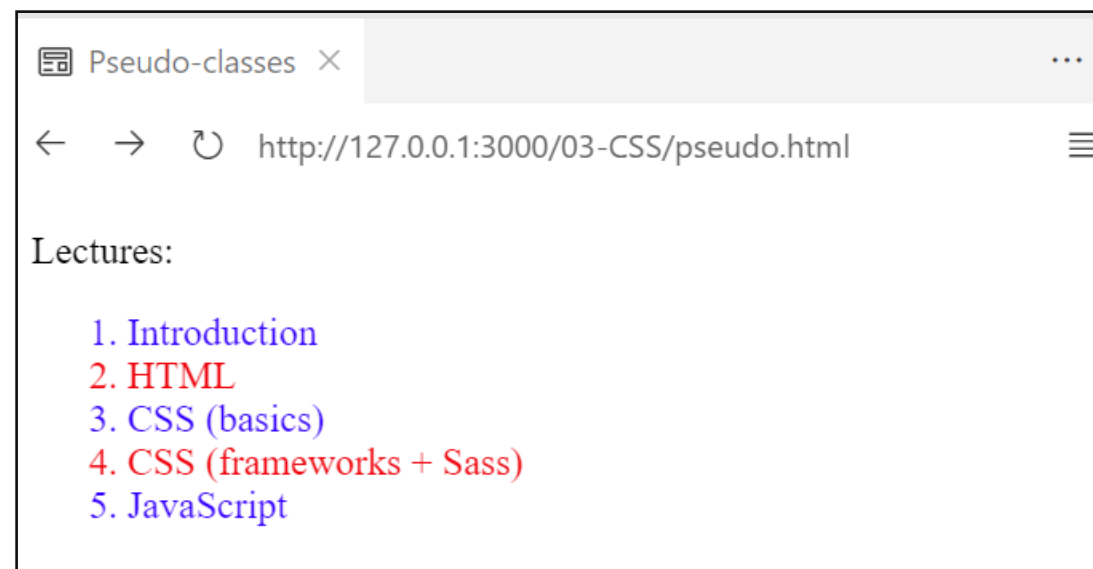
```
em:nth-child(2) {  
  color: red;  
}
```

Ann **Bob** Carl
Ann **Bob** Car

PSEUDO-CLASSES: POSITION RELATIONS

```
<p>Lectures:</p>
<ol>
  <li>Introduction</li>
  <li>HTML</li>
  <li>CSS (basics)</li>
  <li>CSS (frameworks + Sass)</li>
  <li>JavaScript</li>
</ol>
```

```
li:nth-child(even){
  color: red;
}
li:nth-child(odd){
  color: blue;
}
```



PSEUDO-ELEMENTS

Pseudo-elements can be used to target specific parts of the content of a given HTML element, without adding extra HTML markup

The syntax of pseudo-element selectors is **selector::*pseudo-element***

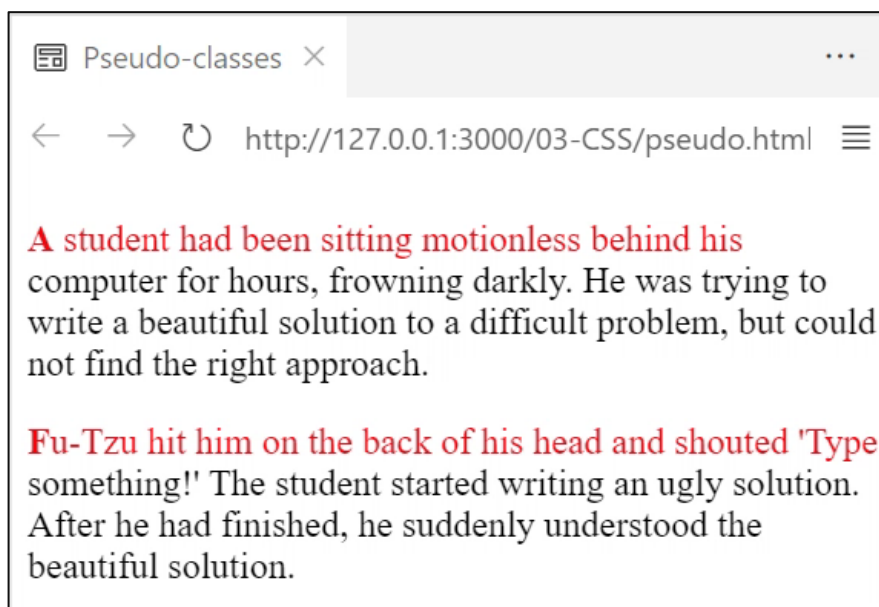
- **selector** is a CSS selector for the target element
- ***pseudo-element*** is one of the supported pseudo-element selectors:
 - **::first-letter**: targets the first letter of the content of a **block-level** element
 - **::first-line**: targets the first line of the content of a **block-level** element
 - **::selection**: targets the content that is currently selected by the user
 - **::before**: creates an element that is the **first child** of the selected element
 - **::after**: creates an element that is the **last child** of the selected element

PSEUDO-ELEMENTS: EXAMPLES

`<p>`A student had been sitting motionless behind his computer for hours, frowning darkly. He was trying to write a beautiful solution to a difficult problem, but could not find the right approach.`</p>`

`<p>`Fu-Tzu hit him on the back of his head and shouted 'Type something!' The student started writing an ugly solution. After he had finished, he suddenly understood the beautiful solution.`</p>`

```
p::first-letter {  
  font-weight:bold;  
}  
p::first-line{  
  color: red;  
}  
p:last-child::selection {  
  background: red;  
  color: white;  
}
```



PSEUDO-ELEMENTS: EXAMPLES

```
<p class="narrator">A hermit spent ten years writing the perfect program. He proudly announced: </p>
<p class="hermit">'My program can compute the motion of the stars on a 286-computer running MS DOS'.</p>
<p class="narrator">Fu-Tzu responded:</p>
<p class="fu-tzu">'Nobody owns a 286-computer or uses MS DOS anymore.'</p>
```

```
.narrator::before {
  content: "Narrator » "; font-weight: bold;
}
.narrator::after {
  content: " «"; font-weight: bold;
}
.hermit::before {
  content: "🧙: "; font-size: 24px;
}
.fu-tzu::before {
  content: "🧙: "; font-size: 24px;
}
```

Pseudo-elements × ...

← → ↺ http://127.0.0.1:3000/03-CSS/pse ☰

Narrator » A hermit spent ten years writing the perfect program. He proudly announced: «

🧙 : 'My program can compute the motion of the stars on a 286-computer running MS DOS'.

Narrator » Fu-Tzu responded: «

🧙 : 'Nobody owns a 286-computer or uses MS DOS anymore.'

THE CASCADE



THE CASCADE IN CASCADING STYLE SHEETS

- Sometimes, two or more rules might apply to the same element
- These rules might be **conflicting**, i.e., assign different values to the same property (e.g.: color)
- **The cascade** is the algorithm used to **resolve** such **conflicts**
 - **Input:** a set of conflicting properties that apply to a given element
 - **Output:** a single, cascaded, property to actually apply
- The cascade considers **4 key aspects, in order:**
 1. **Origin and Importance**
 2. **Layers**
 3. **Specificity**
 4. **Position and order of appearance** of the rule

THE CASCADE: ORIGIN

- The CSS we write (a.k.a. **authored CSS**) is not the only one being applied to a web page
- We've already mentioned that **user agent styles** exist
 - The stylesheets that are included by browsers by default
- Other styles (a.k.a. **local user styles**) might be added by specific browser extensions or from the operating system level
 - For example, for accessibility purposes
 - Visually-impaired persons might want to use high-contrast color schemes, with larger fonts, etc.

THE CASCADE: IMPORTANCE

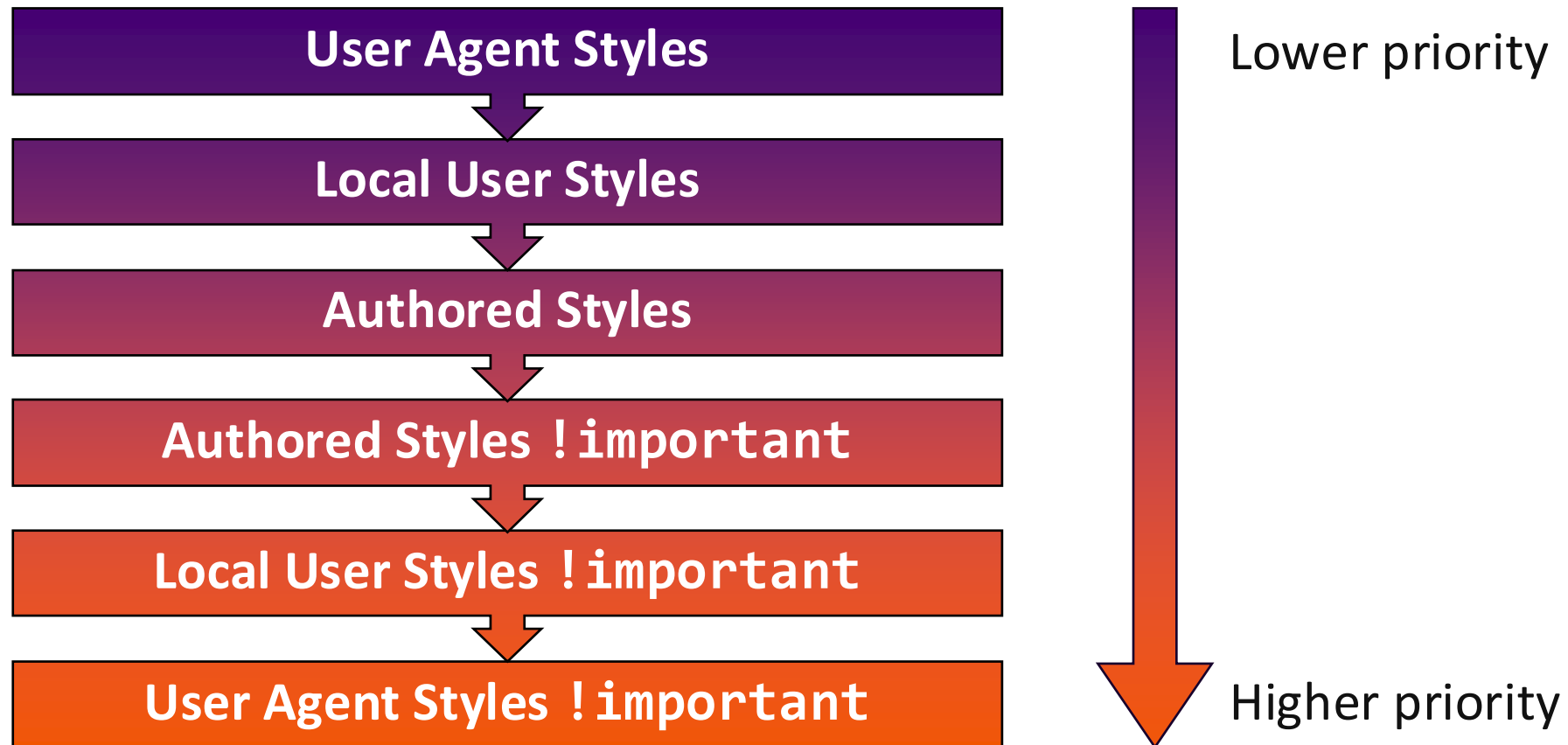
- The `!important` rule can be used to add more importance to a property inside a CSS rule
- `!important` is simply added at the end of the property declaration

```
h1 {  
  color: red !important;  
}
```

- Importance plays a significant role in the cascade

THE CASCADE: ORIGIN-IMPORTANCE

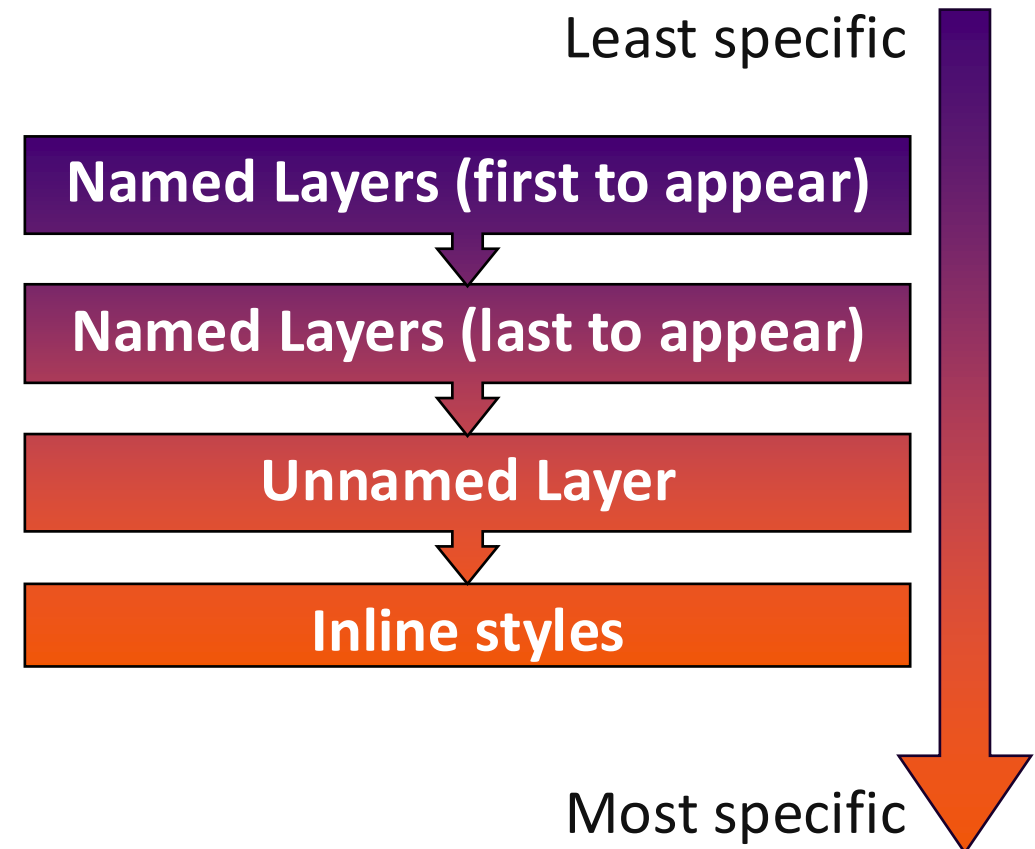
Origins ordered from lowest to highest priority



THE CASCADE: LAYERS

Within each origin/importance bucket, there can be multiple cascade **layers**

- Within authored styles:
 - Custom layers can be defined using @layer rule (we won't see that)
 - The custom layers that are declared later have higher priority
 - All other CSS (in **<style>** or imported with **<link>**) belongs to a unnamed layer
 - Inline styles belong to a separate layer and have the highest priority



THE CASCADE: SPECIFICITY

When two conflicting rules:

- Belong to the same origin-importance bucket, and
- Belong the same layer

Specificity is considered.

- The idea is that the **most specific** selector should win

<pre>.primary { color: blue; }</pre>		<pre>h1 { color: red; }</pre>
--	--	---

<pre><h1 class="primary">Cascading!</h1></pre>
--

THE CASCADE: SPECIFICITY

CSS defines how to calculate the specificity of a selector:

- Ignore the universal selector
- Count the number of id selectors (=A)
- Count the number of class, attribute and pseudo-class selectors (=B)
- Count the number of element and pseudo-element selectors (=C)

The specificity is a numeric triple **(A, B, C)** computed as above

THE CASCADE: SPECIFICITY EXAMPLES

- A: Number of id selectors
- B: Number of class, attribute and pseudo-classes selectors
- C: Number of element and pseudo-element selectors

Selector	Specificity (A, B, C)
<code>#id</code>	(1, 0, 0)
<code>em.master[target]</code>	(0, 2, 1)
<code>#navbar ul li a.nav-link[href*='/']</code>	(1, 2, 3)
<code>article.item section p::first-letter</code>	(0, 1, 4)
<code>a:hover</code>	(0, 1, 1)
<code>*</code>	(0, 0, 0)

THE CASCADE: COMPARING SPECIFICITIES

Comparisons are made by considering the three components in order:

- the specificity with a larger **A** is more specific;
- if the two **A** are tied, then the specificity with a larger **B** wins;
- if the two **B** are also tied, then the specificity with a larger **C** wins;
- if all the values are tied, the two specificities are **equal**.

THE CASCADE: SPECIFICITY



Selector #1	Specif. #1	Selector #2	Specif. #2	Winner
<code>a[target]</code>	(0, 1, 1)	<code>.list a</code>	(0, 1, 1)	Draw
<code>#msg</code>	(1, 0, 0)	<code>input[type].inp</code>	(0, 2, 1)	#1
<code>#nav > #brd a.lk</code>	(2, 1, 1)	<code>em.foo.bar.light</code>	(0, 3, 1)	#1
<code>[id='nav'] a</code>	(0, 1, 1)	<code>#nav a</code>	(1, 0, 1)	#2

THE CASCADE: POSITION AND APPEARENCE

When two properties:

- Belong to the same origin/importance bucket
- Belong to the same layer
- Have the same specificity

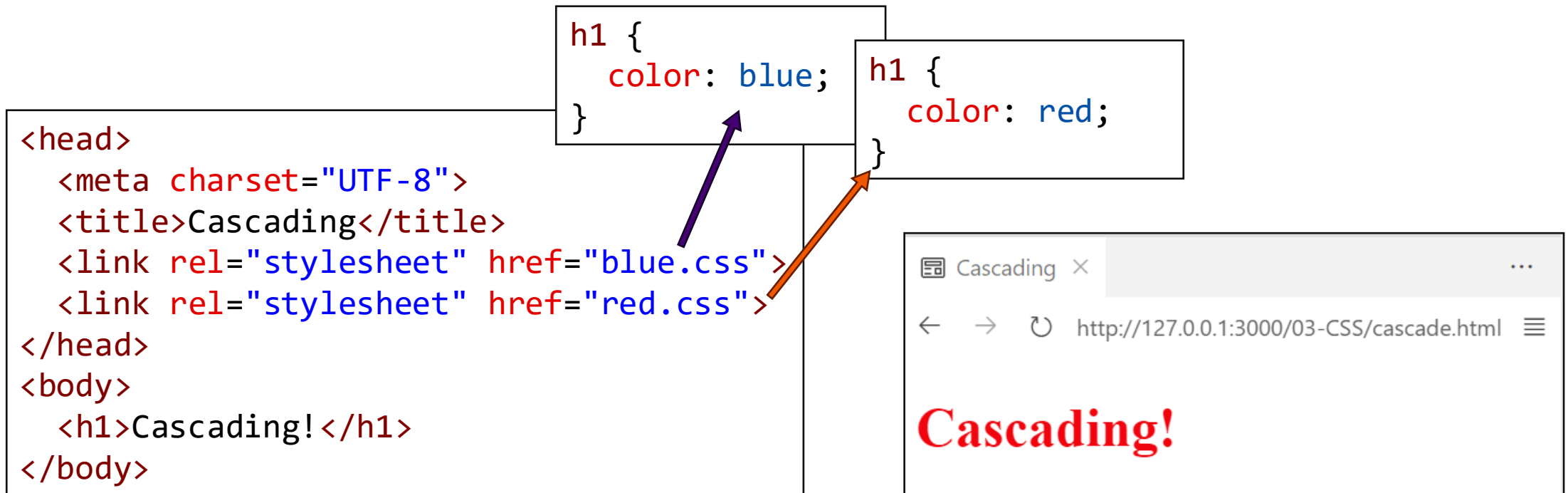
The **last rule** to appear has the highest priority

```
h1 {  
  color: red;  
}  
h1 {  
  color: blue;  
}
```

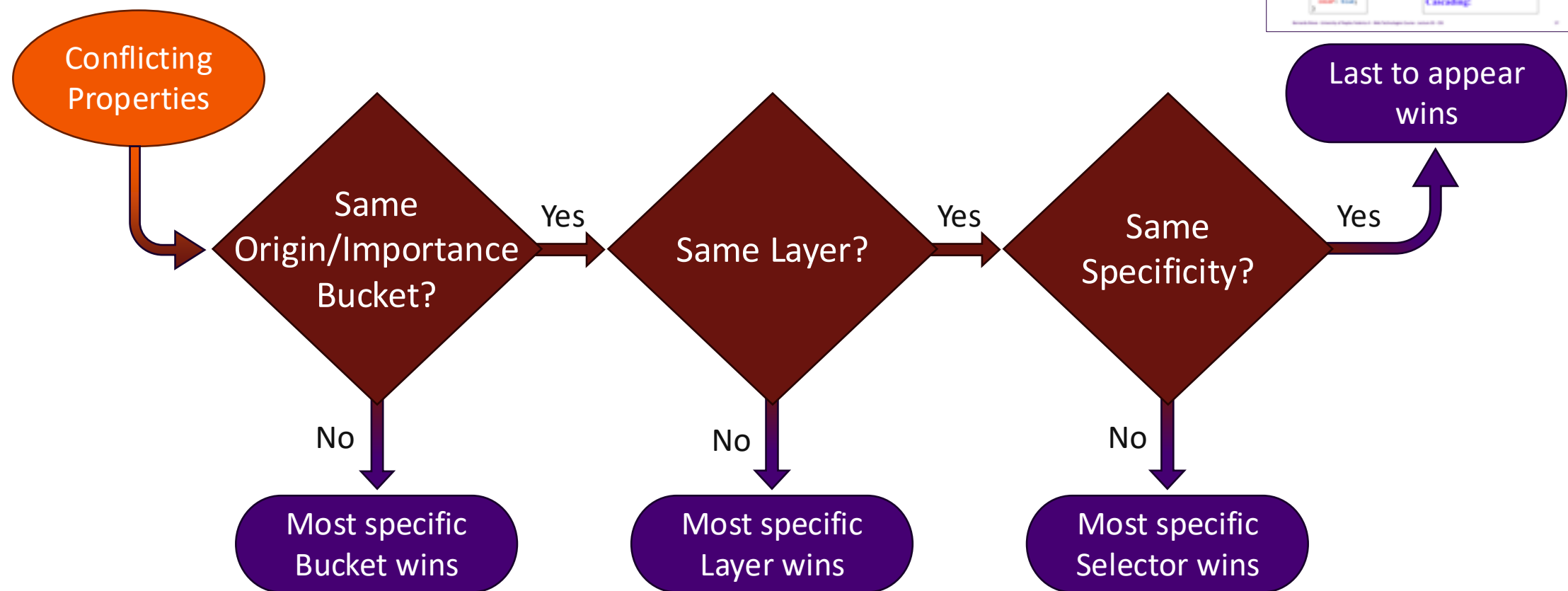


THE CASCADE: POSITION AND APPEARANCE

- This rule applies within the same stylesheet, and on the order in which stylesheets appear



THE CASCADE: OVERVIEW



THE CASCADE: POSITION AND APPEARANCE

When two properties:

- Belong to the same origin/importance bucket
- Belong to the same layer
- Have the same specificity

The last rule to appear has the highest priority

Source: [MDN Web Docs: Cascading Style Sheets \(CSS\)](#)

THE CASCADE: ORIGIN-IMPORTANCE

Origins ordered from lowest to highest priority

Lower priority

Higher priority

Source: [MDN Web Docs: Cascading Style Sheets \(CSS\)](#)

THE CASCADE: LAYERS

Within each origin/importance bucket, there can be multiple cascade layers

- Within authored styles:
 - Custom layers can be defined using `@layer` rule (see [here](#))
 - The custom layers that are declared later have higher priority
- All other CSS (in `<style>` or imported with `<link>`) belongs to a unnamed layer
- Inline styles belong to a separate layer and have the highest priority

Lowest specificity

Highest specificity

Source: [MDN Web Docs: Cascading Style Sheets \(CSS\)](#)

THE CASCADE: SPECIFICITY

CSS defines how to calculate the specificity of a selector:

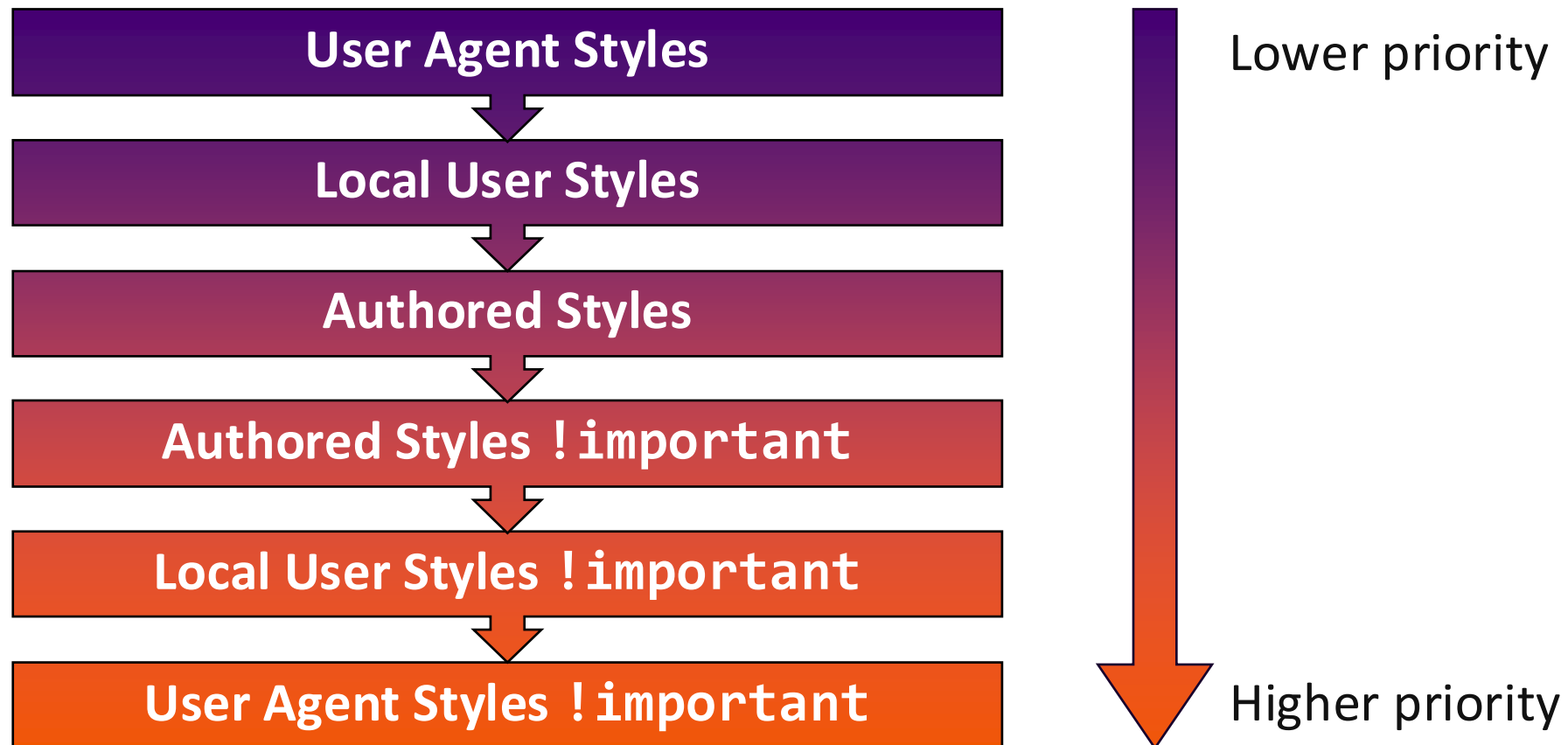
- Ignore the universal selector
- Count the number of id selectors (xI)
- Count the number of class, attribute and pseudo-class selectors (xR)
- Count the number of type and pseudo-element selectors (xC)

The specificity is a numeric triple [A, R, C] computed as above

Source: [MDN Web Docs: Cascading Style Sheets \(CSS\)](#)

THE CASCADE: ORIGIN-IMPORTANCE

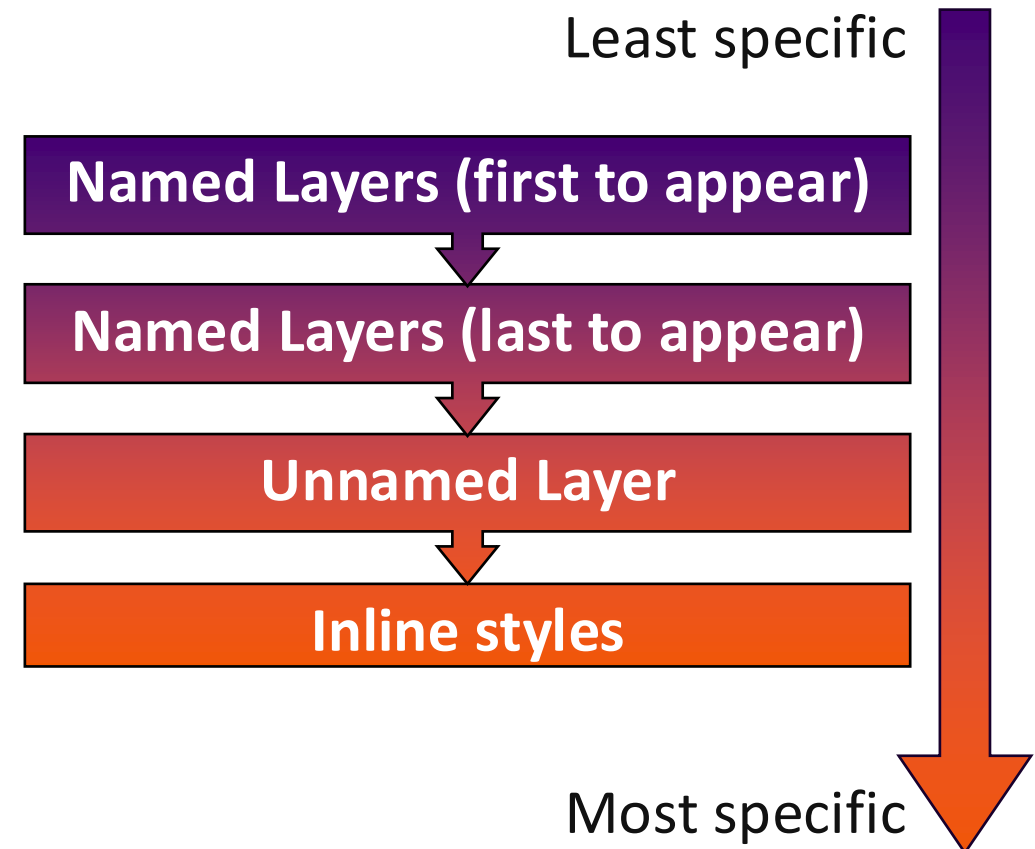
Origins ordered from lowest to highest priority



THE CASCADE: LAYERS

Within each origin/importance bucket, there can be multiple cascade **layers**

- Within authored styles:
 - Custom layers can be defined using @layer rule (we won't see that)
 - The custom layers that are declared later have higher priority
 - All other CSS (in **<style>** or imported with **<link>**) belongs to a unnamed layer
 - Inline styles belong to a separate layer and have the highest priority



THE CASCADE: SPECIFICITY

CSS defines how to calculate the specificity of a selector:

- Ignore the universal selector
- Count the number of id selectors (=A)
- Count the number of class, attribute and pseudo-classes selectors (=B)
- Count the number of type and pseudo-element selectors (=C)

The specificity is a numeric triple **(A, B, C)** computed as above

THE CASCADE: POSITION AND APPEARENCE

When two properties:

- Belong to the same origin/importance bucket
- Belong to the same layer
- Have the same specificity

The **last rule** to appear has the highest priority

```
h1 {  
  color: red;  
}  
h1 {  
  color: blue;  
}
```



THE CASCADE IN BROWSER DEV TOOLS

Cascading!

Inspector Console Debugger Network Style Editor Performance

Search HTML + Filter Styles :hov .cls +

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Cascading</title>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <h1 class="primary">Cascading!</h1>
  </body>
</html>
```

element :: { inline}

h1.primary :: { style.css:1
color: teal;

.primary :: { style.css:4
color: blue;

h1 :: { style.css:7
color: red;

h1 :: { (user agent) html.css:166
display: block;
font-size: 2em;
font-weight: bold;
margin-block-start: .67em;
margin-block-end: .67em;

User agent styles are **hidden** in Dev Tools by default. If you want to see them, press F1 in Dev Tools and change the settings.

Most Specific

Least Specific

INHERITANCE



INHERITANCE IN CSS

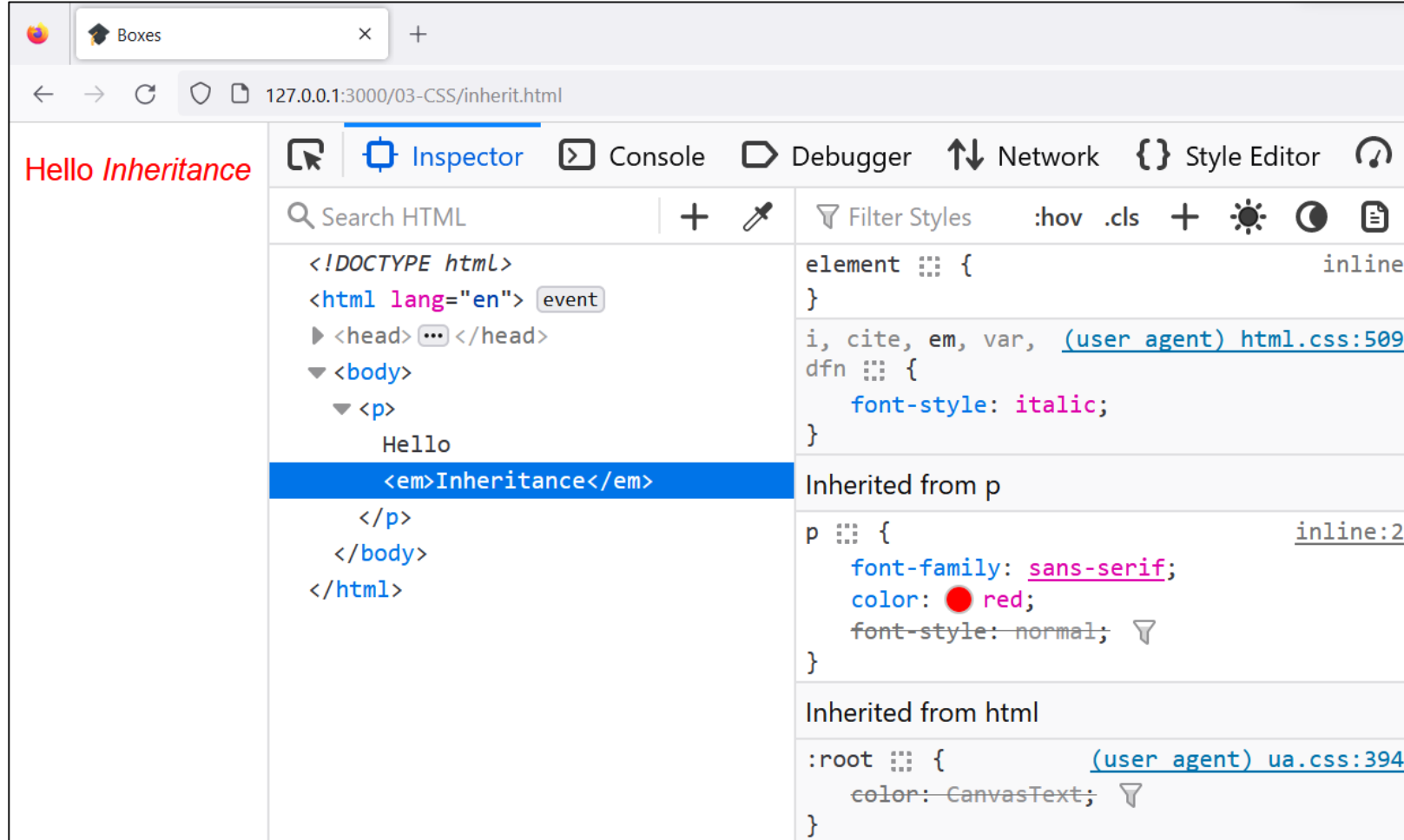
- Some CSS properties can be inherited from ancestor elements, if no specific value is set
- Inheritable properties include **color**, **font-size**, **font-family**, **font-weight**, **font-style**

```
p {  
  font-family: sans-serif;  
  color: red;  
  font-style: normal;  
}
```

```
<p>  
  Hello <em>Inheritance</em>  
</p>
```



INHERITANCE IN CSS



Inherited properties have the lowest specificity of all styling methods

ASSIGNMENT #2

Today's lecture comes with **Assignment #2**! In this assignment, you will:

- Do some practice with basic CSS
- Write some tricky CSS rules
- Test your knowledge of the Cascade algorithm

Note: the live HTTP server we setup in **Exercise 1** of **Assignment #1** will be handy for this assignment! Unless you are already familiar with HTTP servers and already know what you're doing, make sure you completed at least **Exercise 1** in **Assignment #1** before doing **Assignment #2**!

REFERENCES

- **Learn CSS**

web.dev

<https://web.dev/learn/css/>

Sections: 1, 3 to 6, 14, 15

- **Introducing the CSS Cascade**

MDN web docs

<https://developer.mozilla.org/en-US/docs/Web/CSS/Cascade>

- **Flukeout: A game-based approach to learning CSS selectors**

<https://flukeout.github.io/>

